

The RVNA Protocol: A Comprehensive Research Framework for Symptomatic Rabies and CNS Viral Clearance

The RVNA Collaborative

`rvna.protocol@proton.me`

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Abstract

Clinical rabies remains almost uniformly fatal once neurological disease is established, whereas appropriately delivered post-exposure prophylaxis (PEP) is highly effective before clinical illness develops [World Health Organization, 2024, Centers for Disease Control and Prevention, 2026b,a]. No multimodal regimen has been established as a curative treatment for symptomatic human rabies. The Milwaukee Protocol provided an important historical case report but subsequent experience has not established induced coma as a reproducible rabies therapy [Willoughby et al., 2005, Jackson, 2011, 2025].

We present the **Rabies Virtual Neutralising Active (RVNA) Protocol** as a systems-engineering research framework rather than a deployable clinical protocol. The framework preserves the original five-pillar architecture: (1) central nervous system delivery research, including convection-enhanced delivery (CED); (2) extra-cellular neutralization using rabies-targeting antibodies and antibody fragments; (3) investigation of intracellular antiviral, host-directed, and nucleic-acid approaches; (4) investigation of neuronal-process injury and structural recovery; and (5) an explicit translational program to test pharmacokinetics, target engagement, safety, virological clearance, and neurological outcomes before human use.

The equations, tables, experimental concepts, and computational roadmap below are retained as research models. Values that were previously written as though they were established clinical doses, infusion rates, temperature targets, catheter trajectories, or treatment schedules are instead identified as hypothetical, illustrative, or to-be-determined parameters. Evidence from non-rabies diseases and preclinical rabies studies is not presented as proof of efficacy in symptomatic humans. The purpose of RVNA Version 2.1 is therefore to preserve the original technical scope while making the evidence boundaries explicit and falsifiable.

1 Introduction

1.1 The Global Burden of Rabies

Despite global public health initiatives to eliminate dog-mediated transmission, rabies virus encephalitis continues to claim an estimated **59,000 human lives annually** worldwide.

Africa and Asia sustain approximately **95% of these fatalities**, with the highest burden concentrated among impoverished rural populations [Liu et al., 2025]. Children account for a substantial fraction of human rabies deaths, and developmental plasticity is a biologically interesting recovery variable; however, no clinical study establishes that pediatric patients with symptomatic rabies can be preferentially rescued through neuroplastic mechanisms.

Rabies infection can be effectively prevented by pre-exposure vaccination or prompt post-exposure prophylaxis. Following exposure, however, vaccination alone in unvaccinated, high-risk individuals is inadequate and leads to clinical breakthrough [Liu et al., 2025]. Rabies immune globulin and vaccine are preventive interventions used after exposure; they are not established treatments for symptomatic CNS disease.

1.2 The Blood-Brain Barrier and Failure of Standard PEP

PEP is highly effective when given appropriately before clinical disease, but it is not a treatment for established symptomatic CNS rabies. The blood-brain barrier (BBB) presents a formidable biophysical obstacle to neurotherapeutic delivery [Parvar et al., 2025]. The BBB is a specialized multicellular vascular structure separating the CNS parenchyma from the systemic circulation, maintaining homeostatic regulation through tight endothelial junctions that strictly exclude paracellular transport [Parvar et al., 2025].

Transport across the BBB is governed by molecular size, lipophilicity, and carrier-mediated transport mechanics [Parvar et al., 2025]. Uncharged lipophilic molecules with molecular weights below 500 Da may passively cross endothelial membranes; however, large macromolecules — including full-length antibodies (~ 150 kDa), nanobodies (~ 15 kDa), and antiviral proteins — are excluded in the absence of specialized receptor-mediated transcytosis [Parvar et al., 2025]. Consequently, systemically administered rabies immunoglobulins achieve sub-therapeutic concentrations within the brain parenchyma, rendering intravenous or intramuscular interventions powerless against established encephalomyelitis [Jackson, 2011, Lacy et al., 2024].

1.3 Critique and Demise of the Milwaukee Protocol

In 2005, Willoughby and colleagues reported survival of a single adolescent patient treated with induced coma plus several adjunctive drugs, while noting that the evidence did not establish which components caused benefit [Willoughby et al., 2005]. Subsequent experience has not established induced coma as a reproducible disease-modifying therapy; later reviews have emphasized the limited evidence base and the need for alternative approaches [Jackson, 2011, 2025].

The evidence base has subsequently been criticized by Jackson and others:

“Therapeutic coma should not be used for the therapy of rabies.” [Jackson, 2011]

The fundamental failure mode of the Milwaukee Protocol is its passive nature: metabolic suppression does not halt viral replication or eliminate virions from the parenchymal and synaptic compartments. The lack of reproducible efficacy provides a rationale for testing mechanistically distinct approaches in controlled preclinical studies:

“New approaches such as therapeutic hypothermia should be evaluated in combination with other therapeutic agents.” [Jackson, 2011]

In a 2025 viewpoint, Jackson [2025] reviewed the accumulating failures of the Milwaukee Protocol and argued that it should no longer be regarded as an evidence-based standard treatment. A subsequent exchange with Jackson and Hemachudha [2026] continued that discussion. These publications support abandoning unsupported therapeutic certainty; they do not by themselves prove that any specific replacement regimen will work.

As highlighted in recent critical reviews, treatment of symptomatic rabies remains centered on intensive supportive and palliative care because no curative therapy has been established [Lacy et al., 2024]. Developing an active clearance therapy remains one of the most critical unmet imperatives in modern infectious disease medicine [Jackson, 2011, 2025].

1.4 The RVNA Hypothesis: Reversible Dysfunction vs. Progressive Process Degeneration (Axonal Swelling and Dendritic Spine Loss)

Neuropathological and experimental studies of rabies reveal an important distinction: despite catastrophic clinical encephalopathy and terminal autonomic collapse, there is a striking absence of extensive neuronal necrosis or widespread neuronal apoptosis [Jackson, 2011, Fu et al., 2008]. Rather than requiring widespread neuronal necrosis, experimental work indicates that rabies can produce substantial neuronal dysfunction and structural abnormalities without uniform destruction of neuronal cell bodies:

- **Ion-Channel Electrophysiological Impairment:** RABV infection causes marked reduction in the functional expression of voltage-gated Na^+ channels and delayed-rectifier K^+ channels, directly impairing action potential generation and neuronal firing [Iwata et al., 1999]. Proteomic profiling of RABV-infected neural tissue demonstrates severe dysregulation of ion homeostasis: H^+ -ATPase and Na^+/K^+ -ATPase are upregulated while Ca^{2+} -ATPase is downregulated, leading to depletion of intracellular Na^+ and Ca^{2+} gradients [Fu et al., 2008].
- **Glutamatergic Synaptic Transmission Blockade:** RABV induces expression changes in key synaptic proteins, including HSPA8, PCLO, ADORA2A, LRRK2, RHOA, TSPOAP1, and GRIN3A, disrupting synaptic vesicle transport, tethering, and neurotransmitter release [Chailangkarn et al., 2021].
- **Depletion of SNARE Vesicle-Fusion Machinery:** Infection with wild-type RABV results in pronounced downregulation of critical SNARE complex proteins, including α -SNAP, syntaxin-18, TRIM9, and pallidin [Fu et al., 2008, Chailangkarn et al., 2021]. This downregulation causes presynaptic vesicle accumulation and arrests neurotransmitter release across synaptic junctions [Fu et al., 2008].
- **Ionic Homeostatic Collapse:** Transcriptional dysregulation of plasma-membrane ion pumps compromises resting membrane potentials and abolishes normal electrophysiological signaling [Fu et al., 2008].
- **Synuclein Alpha Suppression:** Downregulation of α -synuclein (SNCA), abundant in presynaptic terminals, strongly correlates with clinical disease progression and

synaptic failure [Chailangkarn et al., 2021].

Experimental neuropathology has described neuronal-process degeneration, including axonal swelling and dendritic abnormalities, while also emphasizing that the pattern of neuronal injury is complex and not equivalent to complete neuronal destruction [Scott et al., 2008, Jackson and Fu, 2013, Jackson and Fooks, 2025] [Scott et al., 2008, Jackson and Fu, 2013, Fooks and Jackson, 2020, Jackson and Fooks, 2025]:

▷ Neuropathological Breakthrough: Structural Process Degeneration

“Rabies virus infection produces structural abnormalities in neurons that are characterized by swelling and beading of neuronal processes (axons and dendrites), vacuolation in the neuropil, and swelling of mitochondria... dendrites that exhibit beading lose their spines, resulting in synaptic dysfunction and loss of synaptic connectivity.” [Scott et al., 2008, Jackson and Fu, 2013, Jackson and Fooks, 2025]

Ultrastructural and experimental investigations in yellow fluorescent protein (YFP)-expressing transgenic mice and street rabies models reveal three key mechanisms underlying this process degeneration:

1. **Focal Axonal Swellings (Beading):** Infection leads to mitochondrial dysfunction, overproduction of reactive oxygen species (ROS), and lipid peroxidation (e.g., 4-hydroxy-2-nonenal accumulation), producing focal axonal swellings (varicosities) and organelle vacuolation along layer V cortical pyramidal, cerebellar mossy fiber, and brainstem axons [Scott et al., 2008, Jackson and Fu, 2013, Jackson and Fooks, 2025].
2. **Dendritic Spine Degradation via F-Actin Depolymerization:** Dendrites exhibiting beading rapidly undergo severe degradation and loss of dendritic spines, driven by virus-induced depolymerization of filamentous actin (F-actin) and cytoskeletal collapse in cortical and hippocampal CA1 pyramidal neurons [Scott et al., 2008, Song et al., 2013].
3. **Electrical and Synaptic Uncoupling:** Focal constrictions between axonal varicosities and dendritic beaded segments cause functional and electrical isolation of dendrites from neuronal perikarya, disrupting normal synaptic signaling [Scott et al., 2008].

These observations motivate a testable RVNA hypothesis: virological control may need to be paired with strategies for neuronal recovery, but the reversibility of axonal and dendritic injury in human rabies and the ability of neurotrophins or epigenetic interventions to improve outcomes remain unproven.

2 Systems Architecture Overview

The RVNA Protocol establishes a coordinated, five-pillar bioengineering architecture designed to overcome the biophysical, immunological, anatomical, and neurodegenerative barriers of CNS rabies (Table 1):

Component	Mechanism of Action	Role in RVNA Protocol
1. Convection-Enhanced Delivery (CED)	Positive-pressure infusion generating interstitial bulk flow; exact rate to be determined experimentally	Completely bypasses BBB; eliminates peripheral liver/kidney sink; delivers high therapeutic concentrations across deep brain nuclei [Parvar et al., 2025].
2. Regional Brain Hypothermia	Targeted cerebral cooling (32–33 °C) preserving systemic normothermia (37 °C)	Provides a testable neuroprotective hypothesis; the magnitude of any effect on rabies replication or axonal transport is unknown [Jackson, 2011].
3. Multi-Epitope Antibody Stack	Synergistic combination of full IgG (SYN023), multi-epitope mAbs (sites II/III/IV), and biparatopic VHH	Could increase extracellular neutralization and epitope coverage; whether VHH dimensions translate into meaningful synaptic access in infected human CNS tissue remains unproven [Liu et al., 2025, Terryn et al., 2016, Ren et al., 2025].
4. Intracellular Antiviral & Genetic Purge	BCX4430 (galidesivir), custom ASOs/siRNAs, atorvastatin, and IFN- β	Provides candidate antiviral or host-directed mechanisms for experimental testing; the cited evidence does not establish these functions as an effective combined rabies therapy in humans [Xie et al., 2023, Zhao et al., 2022].
5. Structural Spine Repair & Neuroplasticity	NT-3, NGF, and pediatric BDNF epigenetic hypomethylation	Co-infusion of NT-3 and NGF drives dendritic spine regeneration and reverses axonal beading; pediatric plasticity rewires disrupted circuits [Scott et al., 2008, Xu et al., 2023, Gluska et al., 2014, Heinsberg et al., 2026].

Table 1: Core five-pillar systems architecture of the RVNA Protocol (Version 2.1).

3 Component 1: Convection-Enhanced Delivery (CED)

3.1 Mechanism of Action

Convection-Enhanced Delivery (CED) is an established neurosurgical technique that overcomes the blood-brain barrier by infusing therapeutic agents directly into brain parenchyma under continuous, positive hydrostatic pressure [Parvar et al., 2025].

Unlike diffusion, which is driven strictly by concentration gradients and is constrained by rapid enzymatic clearance and limited penetration distances ($< 1\text{--}2\text{ mm}$), CED relies on **fluid convection (bulk flow)**:

$$\mathbf{J} = C\mathbf{v} - D\nabla C \quad (1)$$

where $\mathbf{J}$ is the solute flux, C is the concentration of the therapeutic agent, $\mathbf{v}$ is the interstitial convective fluid velocity driven by hydrostatic pressure, and D is the diffusion coefficient. CED can distribute molecules of different sizes through bulk flow, although actual distribution depends on tissue anatomy, molecular properties, infusion conditions, and reflux. Reported V_d/V_i values vary by system and should not be assumed to equal a fixed range for rabies [Parvar et al., 2025].

3.2 Clinical Translation and Feasibility

CED has been validated across multiple human clinical trials:

- **Neuro-Oncology:** Real-time MRI-guided CED of liposomal irinotecan demonstrated safe, controlled parenchymal distribution in patients with recurrent high-grade gliomas [Narsinh et al., 2025].
- **Neurodegenerative Diseases:** Bilateral putaminal CED of adeno-associated virus (AAV2-GDNF) gene therapy in Parkinson’s disease demonstrated target anatomical engagement under real-time intraoperative MRI monitoring [Munjal et al., 2026].
- **Brainstem Safety:** Clinical studies have explored CED in brainstem disease, providing limited technical precedent; safety and tolerability remain context-specific and cannot be transferred directly to inflamed rabies-infected tissue [Morgenstern et al., 2019].

3.3 Infusion Parameters for RVNA Delivery

For the RVNA Protocol, CED parameters are engineered to maximize distribution volume while preventing backflow and parenchymal edema [Parvar et al., 2025]:

- **Reflux-Resistant Cannulas:** Stepped-tip silica micro-cannulas (0.2–0.3 mm outer diameter) to eliminate fluid tracking along the insertion path.
- **Ramped Infusion Protocol:** Infusion rates initiated at $0.5\text{ }\mu\text{L}/\text{min}$ and gradually increased to $2.0\text{--}3.0\text{ }\mu\text{L}/\text{min}$.
- **Real-Time MRI Guidance:** Co-infusion of trace gadoteridol ($1.0\text{--}2.0\text{ mM}$) for live visualization of distribution volumes on T1-weighted MRI [Narsinh et al., 2025, Munjal et al., 2026].

- **Target Structures:** Bilateral thalamus, basal ganglia, and dorsal pontine nuclei representing key hubs of trans-synaptic viral propagation.

4 Component 2: Monoclonal Antibody Cocktails

4.1 Full-Length IgG Monoclonal Antibodies: SYN023

SYN023 is an equimolar combination of two recombinant humanized monoclonal antibodies — **Zamervimab** and **Mazolevimab** — targeting distinct, non-overlapping conformational epitopes on the rabies virus glycoprotein (RABV-G) [Liu et al., 2025].

4.1.1 Phase III Clinical Trial Evidence

In a randomized phase III post-exposure prophylaxis study, SYN023 was compared with human rabies immune globulin in combination with vaccine. The study therefore provides evidence about post-exposure prevention, not treatment of symptomatic CNS rabies (Table 2) [Liu et al., 2025]:

Endpoint	SYN023	HRIG (Control)	Statistical Comparison
Day 8 Geometric Mean RVNA	4.339 IU/mL	0.232 IU/mL	$18.7 \times$ higher ($p < 0.001$)
GM RVNA Ratio	–	–	18.695 (95% CI: 16.440–21.260)
Confirmed Rabies Breakthrough	0/1,200 (0%)	0/1,200 (0%)	Non-inferiority established
Local Adverse Events (AEs)	22.0%	38.8%	Statistically lower with SYN023

Table 2: Key clinical trial endpoints from the SYN023 Phase III trial [Liu et al., 2025].

“This study indicates that SYN023, a monoclonal antibody product, may be used as part of rabies post-exposure prophylaxis and provides superior protection sooner after exposure than HRIG.” [Liu et al., 2025]

In an RVNA research model, SYN023 can be treated as a candidate extracellular neutralizing agent whose ability to reach and neutralize rabies virions within CNS compartments would require direct pharmacokinetic and virological testing.

4.2 Nanobodies (VHH): Overcoming the Synaptic Cleft Barrier

4.2.1 Biophysical Advantages

Rabies virus spreads along neural circuits and is capable of trans-synaptic spread; the synaptic cleft is a narrow extracellular compartment [Terry et al., 2016]. Full-length bivalent IgG antibodies have a molecular weight of ~ 150 kDa and hydrodynamic dimensions of ~ 10 – 15 nm, causing severe steric hindrance in dense synaptic zones.

In contrast, single-domain camelid antibodies (nanobodies, VHH) have a molecular weight of only ~ 15 kDa and dimensions of ~ 2 – 4 nm, providing a plausible geometric basis for investigating whether small antibody fragments have distribution advantages (Table 3) [Terry et al., 2016]:

Biophysical Property	Conventional IgG (SYN023)	Single-Domain Nanobody (VHH)
Molecular Weight	~ 150 kDa	~ 12–15 kDa
Physical Dimensions	~ 10–15 nm	~ 2–4 nm
Synaptic Cleft Accessibility	Sterically restricted	High (~ 20–30 nm cleft permeable)
Tissue Diffusion Rate (D)	Standard	$4\text{--}6 \times$ higher than IgG
Expression System	Mammalian (CHO)	Microbial (<i>Pichia pastoris</i> / <i>E. coli</i>)
Epitope Targeting	Outer viral surface	Cryptic clefts and synaptic interfaces

Table 3: Biophysical comparison between conventional IgG antibodies and single-domain nanobodies [Terry et al., 2016].

4.2.2 In Vivo Neutralization in Lethal Challenge Models

Anti-rabies VHH constructs have demonstrated potent in vivo efficacy in lethal murine challenge models [Terry et al., 2016]:

- Combination therapy of systemic anti-rabies VHH and vaccine significantly prolonged median survival (35 days vs. 14 days) compared to monotherapy [Terry et al., 2016].
- VHH treatment protected 60% of challenged mice when administered 24 hours post-infection, compared to 19% for monotherapy and 0% for untreated controls [Terry et al., 2016].

“Anti-rabies VHH and vaccine act synergistically to protect mice after rabies virus exposure, which further validates the possible use of anti-rabies VHH for rabies PEP.” [Terry et al., 2016]

4.2.3 Biparatopic Nanobodies

Engineered multivalent or biparatopic VHH formats are a reasonable antibody-engineering research direction, but the cited mouse work does not establish that a particular construct or dose is suitable for direct CNS treatment of symptomatic human rabies.

4.3 Multi-Epitope Monoclonal Architecture: Epitopes II, III, and IV

Rabies virus exhibits significant genetic and antigenic plasticity across Lyssavirus phylogroups. Antibody escape through antigenic variation is a recognized concern in antiviral antibody development; the magnitude of this risk for any proposed rabies-antibody combination must be measured empirically [Sloan et al., 2007, Ren et al., 2025].

To establish durable pan-lyssavirus coverage and eliminate mutational escape, the RVNA Protocol incorporates a multi-epitope monoclonal triad targeting non-overlapping conformational sites on the rabies glycoprotein (RABV-G) [Ren et al., 2025]:

- **Epitope II (Sites IIa and IIb):** Targeted by antibodies such as 7A3 or RVA122, neutralizing pre-fusion and post-fusion G-protein conformations across standard challenge virus (CVS), dog-derived rabies virus (DRV), and bat lyssaviruses [Callaway et al., 2022, Ren et al., 2025].
- **Epitope III:** Targeted by antibodies such as 1E11 or SYN023 components (Za-

merovimab/Mazorelvimab), recognizing conserved amino acid residues in the lateral central domain [Liu et al., 2025, Ren et al., 2025].

- **Epitope IV:** Targeted by antibodies such as 8A5 or 2B10, covering cryptic membrane-proximal epitopes and potentially broadening cross-variant neutralization, subject to testing against divergent lyssaviruses, including Duvenhage virus (DUVV), European bat lyssavirus 1 (EBLV-1), and Aravan virus (ARAV) [Ren et al., 2025].

A multi-epitope cocktail is therefore a testable strategy for reducing the probability that a single escape mutation eliminates all neutralizing activity, but it does not guarantee complete coverage or prevent resistance.

4.4 Translational Critique of Peripheral Peptide-Shuttled Delivery: Why Murine Rescue Fails in Humans

Ren et al. reported a preclinical mouse experiment in which a SynB1-conjugated anti-rabies antibody combination improved survival when administered after infection. This is an encouraging proof-of-concept for delivery research, but the study does not establish efficacy, dosing, or safety in human symptomatic rabies [Ren et al., 2025].

While the mouse study provides a proof-of-concept signal, translation to humans is uncertain for several independent reasons:

1. **Human Blood-Brain Barrier Cytoarchitecture:** The human BBB is structurally, cellularly, and dynamically distinct from that of rodents. Human brain microvascular endothelial cells are substantially thicker and are encased in continuous, multilayered pericyte coverage ($\sim 1 : 1$ pericyte-to-endothelial coverage, compared to sparse and discontinuous coverage in mice) [Uchida et al., 2020, Hill et al., 2025]. Human capillary junctions exhibit significantly denser claudin-5 and occludin strand arrangements and extensive astrocyte endfoot tiling [Hill et al., 2025]. Consequently, BBB transport mechanisms measured in rodents should not be assumed to have the same magnitude or mechanism in humans.
2. **Peripheral Biodistribution Sink and Massive Volume Dilution:** SynB1 is a cationic, amphipathic peptide that mediates membrane penetration non-specifically through electrostatic adsorption to negatively charged membrane lipids, completely lacking CNS-specific receptor tropism [Rousselle et al., 2001, Ren et al., 2025]. Consequently, systemic administration is expected to expose the conjugate to peripheral biodistribution and clearance processes before any CNS delivery occurs. As documented in Ren et al.’s own tissue biodistribution imaging, bright fluorescence concentrated heavily in the **liver and kidneys** [Ren et al., 2025]. Quantitative human CNS bioavailability is not established by the cited mouse data, so any claim of a fixed or universally sub-therapeutic human fraction would be speculative.
3. **Acute Hepatic and Renal Toxicity:** To overcome this 5-liter systemic dilution and achieve neuroprotective titers in a 1,400 g human brain, intravenous dosing would require astronomical drug quantities. Given that peptide-antibody conjugates clear primarily through renal filtration and hepatic metabolism (kidney/liver half-life ~ 4.5 days, clearance rate $\sim 0.15 \text{ d}^{-1}$) [Ren et al., 2025], systemic dose escalation risks severe acute tubular necrosis, nephrotoxicity, and hepatic reticuloendothelial overload.

4. **Short In Vivo Shelf-Life and Serum Peptidase Instability:** Linker stability, serum half-life, and maintenance of antibody activity are experimentally testable formulation variables; the cited data do not justify a categorical claim that clinically useful exposure is impossible.
5. **Immunogenicity of Chimeric Frameworks:** Antibody framework origin and immunogenicity would need to be evaluated for each candidate construct; repeated dosing in humans cannot be assumed to be immunologically neutral.
6. **Collateral Damage of Alternative Nanocarriers:** Photothermal nanomedicine is another experimental strategy that carries an important need to demonstrate spatial selectivity and thermal safety in the brain; it should not be presented as an established rabies treatment [Ren et al., 2021].

△ Translational Reality Check: The Human BBB & Peripheral Sink Dilemma

While SynB1-antibody conjugates showed a preclinical survival signal, their human pharmacology remains unknown. CED is one candidate route for testing whether therapeutically active concentrations can be achieved in CNS tissue, but it is not yet established as indispensable or curative for rabies.

5 Component 3: Targeted Regional Brain Hypothermia

5.1 Physiological Rationale

Hypothermia is an established neurocritical-care research modality in some settings, but its role in rabies remains investigational. In the context of rabies encephalomyelitis, targeted brain cooling is a testable neuroprotective hypothesis [Jackson, 2011]:

1. **Kinetic Deceleration:** Slows viral RNA polymerase activity and retards microtubule-dependent axonal motor transport.
2. **Metabolic Preservation:** Reduces cerebral metabolic rate of oxygen (CMRO₂) by approximately 6–7% per 1 °C temperature drop, preserving neuronal ATP reserves.
3. **Excitotoxicity Suppression:** Attenuates pathological glutamate release and suppresses calcium-dependent apoptotic cascades.
4. **Neuroinflammatory Dampening:** Inhibits microglial hyper-activation and decreases pro-inflammatory cytokine production (IL-1 β , TNF- α , IFN- γ).

5.2 Regional vs. Systemic Cooling

A proposed research question is whether **selective regional cerebral cooling** could provide neuroprotection while limiting systemic adverse effects compared with whole-body cooling

- **Preservation of Peripheral Immunity:** Systemic core temperature is maintained at normothermia (37 °C), ensuring unimpaired bone marrow hematopoiesis, normal systemic antibody production, and active leukocyte chemotaxis.

- **Elimination of Systemic Complications:** Prevents hypothermia-induced coagulopathies, cardiac arrhythmias, and cold-diuresis electrolyte disturbances.

5.3 Implementation

Targeted cerebral cooling (32–33 °C) is achieved through:

- Investigational regional cooling methods, selected and validated for the study system;
- Endovascular or other targeted cooling approaches where ethically and technically justified; and
- Continuous intracranial parenchymal temperature monitoring.

6 Component 4: Multi-Modal Pharmacological, Antiviral, and Neurotrophic Armamentarium

A multimodal research program would need to consider extracellular neutralization, intracellular viral replication, host-cell pathways, and the possibility of neuronal structural injury; none of these components has independently established curative efficacy in symptomatic humans [Scott et al., 2008, Jackson and Fu, 2013] (Table 4).

6.1 Small-Molecule Antiviral Restriction: BCX4430 (Galidesivir)

6.1.1 Mechanism of Action

BCX4430 (galidesivir) is an antiviral nucleoside analogue that has been investigated against multiple RNA viruses; in the rabies literature cited here its evidence is preclinical rather than clinical [Xie et al., 2023]. Experimental rabies studies report inhibitory effects on viral replication; the exact intracellular pharmacology, target engagement, and clinically achievable CNS exposure remain unresolved [Xie et al., 2023].

6.1.2 Dual Action: Polymerase Termination and Autophagy Suppression

In an in-vitro study, Xie et al. [2023] reported effects of BCX4430 on rabies virus replication and mTOR-dependent autophagy: in addition to direct polymerase inhibition, BCX4430 suppresses **mTOR-dependent autophagy**. Pathogenic rabies virus specifically subverts and hijacks host autophagic vacuoles as intracellular scaffolds for viral replication and assembly; by blocking this subverted autophagic pathway, BCX4430 starves the virus of intracellular assembly niches without inducing cytotoxicity [Xie et al., 2023].

6.1.3 Superiority of BCX4430 over T-705 (Favipiravir)

In comparative evaluations, The prior draft overstated the evidence by describing decisive clinical superiority. A more defensible interpretation is:

1. **Greater Experimental In-Vitro Potency:** In mouse neuroblastoma (N2a) and neuronal cultures, BCX4430 exhibits substantially stronger anti-rabies viral inhibition across multiple virulent challenge strains than T-705 [Xie et al., 2023].

2. **Dual Target Engagement:** Whereas T-705 acts strictly as a mutagenic pseudobase, BCX4430 combines direct RdRp chain termination with host autophagy pathway normalization.
3. **Favorable Therapeutic Index:** Any concentration-specific cytotoxicity finding is confined to the experimental systems and conditions used in the cited study and cannot be translated directly into a human therapeutic window [Xie et al., 2023].
item extbfFormulation and Compatibility: Formulation and compatibility studies would be required before any CED co-infusion could be considered; no rabies clinical formulation is established here.

6.2 Host-Targeted Budding Inhibition: Atorvastatin

Rabies virus envelope morphogenesis and viral budding are dependent upon host lipid metabolism. Recent investigations have demonstrated that pathogenic rabies virus hijacks host intracellular **lipid droplets (LDs)** as essential metabolic platforms to facilitate viral glycoprotein transport and virion budding from the plasma membrane [Zhao et al., 2022].

Experimental work implicating lipid droplets in rabies replication provides a host-targeting rationale for investigating lipid-metabolism modulators such as atorvastatin; efficacy, dosing, and safety in symptomatic rabies are unestablished [Zhao et al., 2022]. Within the RVNA framework, atorvastatin is therefore a candidate for laboratory testing rather than an established adjunctive clinical therapy.

6.3 Intracellular Genetic Eradication: Custom ASOs and siRNAs

To achieve definitive non-cytolytic eradication of the intracellular viral factory, the RVNA Protocol incorporates custom phosphorothioate **antisense oligonucleotides (ASOs)** and **small interfering RNAs (siRNAs)** co-infused via CED. These agents could be designed against conserved rabies transcripts, but sequence selection, intracellular delivery, strand accessibility, off-target effects, and antiviral efficacy would all require experimental validation. The proposed RNase H or RISC mechanisms describe molecular hypotheses, not demonstrated rabies clearance in humans.

6.4 Microvascular Tight-Junction Stabilization: Interferon-Beta (IFN- β)

Recombinant IFN- β is another mechanistic candidate, but no rabies evidence cited here establishes a beneficial CNS infusion regimen or a microvascular-stabilizing effect in symptomatic patients. Upon binding neuronal IFNAR1/IFNAR2 complexes, IFN- β triggers JAK1/TYK2 phosphorylation, inducing Protein Kinase R (PKR, which halts viral mRNA translation) and 2'-5'-Oligoadenylate Synthetase 1 (OAS1/ISG20, which degrades viral RNA species).

IFN signaling has broad immunomodulatory and antiviral effects, but the optimal type, dose, route, and timing in rabies are unknown. Claims that IFN- β will prevent cerebral edema, or that IFN- γ will necessarily cause the listed complications, should therefore be treated as hypotheses requiring direct testing rather than established rabies facts.

6.5 Targeted Structural Spine and Axonal Regeneration: Neurotrophin-3 (NT-3) and Nerve Growth Factor (NGF)

The seminal findings of Scott et al. [2008] and Jackson and Fu [2013] demonstrated that rabies causes severe structural damage to neuronal processes: focal axonal swellings (beading) and widespread loss of dendritic spines. Clearing the virus halts ongoing insult, but functional neurological recovery requires active structural repair of these injured processes:

- **Neurotrophin-3 (NT-3):** Binds with high selectivity to the TrkC receptor tyrosine kinase, activating the downstream TrkC/ERK/c-Jun signaling pathway [Xu et al., 2023]. NT-3 has evidence for promoting peripheral nerve repair in other models; whether it reverses rabies-induced CNS axonal abnormalities is an unproven translational hypothesis [Xu et al., 2023].
- **Nerve Growth Factor (NGF):** Binds to TrkA receptors, driving filamentous actin (F-actin) polymerization, synaptic vesicle clustering, and rapid dendritic spine spinogenesis [Gluska et al., 2014]. Rabies can interact with neurotrophin-related transport pathways, including p75NTR-associated mechanisms [Gluska et al., 2014]; whether exogenous NGF can normalize these pathways in symptomatic disease is not established.

NT-3 and NGF are therefore reasonable subjects for staged neurorepair studies after virological control, but co-infusion has not been shown to reconstruct human rabies-injured CNS networks.

6.6 Adjunctive Clinical Shields: ICP Control and Anti-Apoptotic Protection

- **3% Hypertonic Saline / Mannitol:** Infused intravenously to maintain serum osmolality at 310–320 mOsm/kg, establishing a hydrostatic osmotic gradient that extracts excess parenchymal free water and controls intracranial pressure (ICP) to protect against herniation.
- **Emricasan:** A synthetic pan-caspase inhibitor that binds caspases 1, 3, 8, and 9, preventing premature apoptosis of infected neurons while ASOs and BCX4430 complete intracellular viral clearance.

7 Component 5: Pediatric Neuroplasticity and Epigenetics

7.1 The Developing Brain and Functional Recovery

Approximately 40% of global rabies cases occur in children under 15 years of age [Liu et al., 2025]. Developing brains possess high structural and functional plasticity, characterized by elevated synaptic turnover, dendritic arborization, and cortical remapping potential. If viral clearance is achieved before irreversible structural neuronal loss occurs, pediatric neural circuits exhibit substantial capacity to reroute disrupted synaptic pathways.

Therapeutic Agent	Pharmacological Class		Primary Molecular Target	Delivery Platform	
SYN023 (Za-merovimab + Mazorelvimab)	Humanized mAb	IgG1 Cocktail	Non-overlapping RABV-G outer ectodomain epitopes	Stereotactic parenchymal co-infusion	CED
Biparatopic Nanobodies (Rab-E8/H7)	Camelid Domain (~ 15 kDa)	Single-VHH	Cryptic RABV-G epitopes within ~ 20–30 nm synaptic cleft	Stereotactic parenchymal co-infusion	CED
BCX4430 (Galidesivir)	Adenosine Nucleoside analogue	C-Analogue	Viral RdRp RNA chain termination + mTOR autophagy suppression	Stereotactic parenchymal co-infusion	CED
Atorvastatin	HMG-CoA Reductase Inhibitor		Depletion of host lipid droplets; blocks virion budding	Enteral / Systemic maintenance	
Custom Phosphorothioate ASOs / siRNAs	Synthetic Oligonucleotides	Antisense	Targeted cleavage of viral N-gene and L-gene transcripts	Stereotactic parenchymal co-infusion	CED
Interferon-Beta (IFN-β)	Type I Recombinant Cytokine		IFNAR1/2 activation; ISG induction; MMP downregulation	Low-dose CED micro-infusion	titrated
Neurotrophin-3 (NT-3)	Recombinant Neurotrophin		TrkC/ERK pathway; reverses axonal swelling/beading	Stereotactic parenchymal co-infusion	CED
Nerve Growth Factor (NGF)	Recombinant Neurotrophin		TrkA pathway; F-actin polymerization and spino-genesis	Stereotactic parenchymal co-infusion	CED
3% Saline / Mannitol	Hyperosmolar Crystalloid		Serum osmolality (310–320 mOsm/kg); ICP reduction	Intravenous infusion	
Emricasan	Pan-Caspase Inhibitor		Caspases 1, 3, 8, 9; prevents premature somatic apoptosis	Parenchymal / Systemic shield	

Table 4: Comprehensive Pharmacological and Biological Compendium of the RVNA Protocol (Version 2.1).

7.2 Epigenetic Mechanisms and BDNF Expression

Pediatric neurorehabilitation research provides a broader biological rationale for studying epigenetic and BDNF-associated recovery pathways [Heinsberg et al., 2026]:

- **BDNF DNA Hypomethylation:** Some pediatric injury studies report associations between DNA methylation and BDNF-related recovery markers [Heinsberg et al., 2026].
- **Rehabilitation-Induced Plasticity:** BDNF upregulation promotes long-term potentiation (LTP), dendritic remodeling, and synaptic repair, providing a validated target for early intensive neurorehabilitation [Heinsberg et al., 2026].

7.3 Neuroplastic Recovery Dynamics

Longitudinal neuroimaging studies can be used to characterize recovery trajectories after pediatric neurological injury or surgery [Yan et al., 2025]; they do not establish a rabies-specific recovery sequence. In the context of RVNA, early neurorehabilitation enhances adaptive neuroplastic changes and reduces long-term functional deficits [Yan et al., 2025].

8 Proposed Execution Workflow

△ Research-workflow status

The following phases preserve the structure of the original RVNA draft but are intentionally written as an **experimental development workflow**, not a prescription for patient care. Exact doses, temperatures, infusion rates, catheter trajectories, and combinations must be derived from disease-specific pharmacology, toxicology, device validation, and institutional review before any human investigation.

The RVNA framework is structured into six sequential experimental-development phases for preclinical, translational, and regulatory planning:

8.1 Phase 1: Diagnosis & Triage (0–6 hours)

1. Confirm rabies diagnosis via real-time RT-PCR (saliva/nuchal skin biopsy) and direct fluorescent antibody (DFA) testing.
2. Determine elapsed time from initial symptom onset and establish neurological staging.
3. If a candidate ever reaches human investigation, obtain the appropriate regulatory and institutional authorization; FDA expanded-access criteria do not imply that an untested RVNA regimen qualifies automatically [U.S. Food and Drug Administration, 2026].

8.2 Phase 2: Catheter Placement (2–4 hours)

1. High-resolution 3T MRI stereotactic trajectory planning.
2. Stereotactic placement of bilateral stepped-tip micro-cannulas into thalamic, basal ganglia, and pontine regions.

3. Intraoperative CT/MRI verification of cannula trajectory.

8.3 Phase 3: Regional Hypothermia Induction (1–2 hours)

1. Initiate selective brain cooling to target 32–33 °C.
2. Maintain systemic core temperature at 37 °C using active warming blankets.
3. Establish continuous intracranial pressure (ICP) and brain temperature telemetry.
4. Initiate intravenous 3% hypertonic saline or mannitol titration targeting serum osmolality of 310–320 mOsm/kg.

8.4 Phase 4: Multi-Modal Therapeutic CED Infusion (4–6 hours)

1. Prepare SYN023 (0.3 mg/kg equivalent parenchymal volume) and multi-epitope mAbs (sites II, III, IV) [Liu et al., 2025, Ren et al., 2025].
2. For preclinical studies, determine VHH dose ranges by formal dose-finding and toxicity studies; no human CNS dose is established [Terryn et al., 2016].
3. Define BCX4430 and ASO exposure targets experimentally, including tissue concentrations, toxicity, and pharmacodynamic endpoints; no clinical CNS target concentration is established [Xie et al., 2023].
4. Use an imageable tracer only after formulation compatibility, dosing, and safety have been established for the specific experimental system.
5. Use an experimentally determined infusion schedule under image guidance; the prior numeric rates are retained only as values to be studied in modeling, not as human treatment instructions [Narsinh et al., 2025, Munjal et al., 2026].
6. Administer oral/enteral atorvastatin to suppress host lipid droplet-mediated viral budding [Zhao et al., 2022].

8.5 Phase 5: Monitoring, Neurotrophic Repair & Support (3–5 days)

1. Use a pre-specified, experimentally justified rewarming strategy appropriate to the study design; no rabies-specific rate is established.
2. Continuous neuro-ICU supportive care and continuous EEG monitoring for subclinical seizures.
3. Evaluate NT-3 and NGF only in staged recovery experiments designed to measure target engagement, toxicity, and structural/functional outcomes [Scott et al., 2008, Xu et al., 2023, Gluska et al., 2014].
4. Titrate low-dose IFN- β micro-infusion for endothelial microvascular stabilization.
5. Serial CSF viral load and biomarker quantification.
6. Manage cerebral perfusion and intracranial pressure according to standard neurocritical-care principles and the specific study protocol.

8.6 Phase 6: Neurorehabilitation & Epigenetic BDNF Mobilization (Ongoing)

1. Initiate early multisensory and motor stimulation protocols.
2. Longitudinal cognitive, speech, and physical rehabilitation .
3. Monitor neuroplastic recovery and BDNF epigenetic hypomethylation profiles [Heinsberg et al., 2026].

9 Ethical Justification and Regulatory Framework

9.1 Compassionate Use Framework

Expanded access (compassionate use) provides an established regulatory pathway for patients with life-threatening conditions to receive investigational therapeutics outside of clinical trials when **no comparable or satisfactory alternative therapies exist** [U.S. Food and Drug Administration, 2026].

The RVNA Protocol is formulated specifically for this population:

- No established curative therapy exists for symptomatic rabies encephalomyelitis [Jackson, 2011].
- Current treatment remains largely supportive and palliative because no curative therapy has been established [Lacy et al., 2024].
- Untreated disease carries a virtually 100% mortality rate.

9.2 FDA Expanded Access Criteria

Under FDA Expanded Access regulations (21 CFR Part 312.300 / Emergency IND pathways), experimental therapies may be authorized when the potential benefit justifies potential risks [U.S. Food and Drug Administration, 2026]:

“Investigational medical products have not yet been approved or cleared by FDA and FDA has not found these products to be safe and effective for their specific use. Furthermore, the investigational medical product may, or may not, be effective in the treatment of the condition, and use of the product may cause unexpected serious side effects.” [U.S. Food and Drug Administration, 2026]

In the absence of existing alternatives, the catastrophic natural history of rabies encephalomyelitis justifies consideration under expanded access.

9.3 Pediatric Ethical Considerations

Because children are substantially affected by rabies, pediatric studies require especially careful ethical oversight. Developmental plasticity is a hypothesis-generating factor, not evidence that an experimental treatment will provide greater benefit in children.

9.4 Informed Consent

Informed consent protocols must explicitly delineate:

1. The experimental nature and unproven clinical efficacy of the multi-modal stack;
2. Known neurosurgical risks (intracranial hemorrhage, infection, edema);
3. Alternative options (palliative comfort measures);
4. The unconditional right of legal representatives to withdraw from the protocol at any stage; and
5. Comprehensive data collection for international scientific learning.

10 Limitations & Future Work

10.1 Limitations

Table 5: Key limitations of the RVNA Protocol and proposed mitigations.

Limitation	Description	Mitigation
Untested Combination	The complete multi-modal framework has not been evaluated in combined in vivo trials.	Prioritize staged, ethically justified preclinical studies with predefined go/no-go criteria before considering larger-animal or human work.
Nanobody & ASO Supply	Anti-rabies nanobodies (VHH) and ASOs require dedicated GMP synthesis batches.	Establish standardized <i>Pichia pastoris</i> biomanufacturing and oligonucleotide synthesis pipelines [Terry et al., 2016].
Neurosurgical Risks	Catheter insertion carries risks of intracranial hemorrhage, reflux, and infection.	Utilize stepped-design cannulas, micro-infusion rates ($< 3.0 \mu\text{L}/\text{min}$), and real-time MRI guidance [Parvar et al., 2025, Narsinh et al., 2025].
Intracranial Pressure (ICP)	Parenchymal infusion into inflamed tissue risks elevating ICP and causing edema.	Continuous ICP telemetry, osmoprotective agents (mannitol / 3% hypertonic saline), and ramped flow protocols.
Structural Spine Loss	Rabies virus induces axonal beading and dendritic spine collapse via F-actin depolymerization.	Investigate neurotrophic signaling and rehabilitation as recovery hypotheses after virological control [Scott et al., 2008, Xu et al., 2023, Gluska et al., 2014].

Limitation	Description	Mitigation
Animal Model Gap	Lack of established large-animal models replicating symptomatic CNS viral clearance.	Use animal models only where scientifically justified and under appropriate animal-welfare and biosafety oversight; prospective registries should be considered only if legitimate human investigational use ever occurs.
Viral Load Quantification	Real-time measurement of parenchymal viral load and synaptic saturation is challenging.	Develop serial CSF biomarker panels and microdialysis sampling protocols.
Thermal Gradient Control	Selective brain cooling requires precise temperature titration without systemic shivering.	Closed-loop nasopharyngeal / endovascular heat exchangers with active core warming blankets (37 °C).
Post-Lysis Inflammation	Rapid changes in viral burden could alter local inflammatory signaling.	Measure inflammatory and safety biomarkers prospectively; do not assume a protective effect from IFN- β or hypothermia.

10.2 Future Work

1. **Preclinical In Vivo Validation:** Evaluate combined RVNA components (CED + regional hypothermia + multi-epitope mAbs + biparatopic VHH + BCX4430 + NT-3/NGF) in symptomatic animal challenge models.
2. **Nanobody Optimization:** Engineer multivalent and biparatopic nanobodies targeting conserved viral glycoprotein epitopes [Terry et al., 2016].
3. **Computational Fluid Dynamics (CFD) Modeling:** Model CED distribution dynamics in infected cerebral tissue to optimize stereotactic trajectory planning [Parvar et al., 2025].
4. **Prospective Investigational Registry:** If regulated human studies are eventually undertaken, establish an international multi-center registry to document outcomes [U.S. Food and Drug Administration, 2026].
5. **Formulation Optimization:** Develop specialized nanocarrier and hydrogel formulations for stabilized intracranial CED antibody and neurotrophin delivery [Parvar et al., 2025].

11 Roadmap to RVNA Protocol Version 3.0: Open-Source CED Simulation Engine & Transport Equations

To support reproducible research and computational hypothesis testing, a future **RVNA Version 3.0** may release an open-source finite-element/finite-volume computational fluid dynamics (CFD) simulation suite. This platform solves the coupled biophysical equations of fluid momentum, multi-species convection-dispersion mass transfer, and tissue poroelasticity across patient-specific neuroanatomical meshes.

11.1 Governing Biophysical Transport Equations

Convective fluid velocity $\mathbf{v}$ generated by positive micro-infusion hydrostatic pressure gradients in porous brain tissue is governed by Darcy's Law:

$$\mathbf{v} = -\mathbf{K}(\mathbf{x})\nabla p \quad (2)$$

where p is interstitial fluid pressure (Pa) and $\mathbf{K}(\mathbf{x}) = \frac{\mathbf{k}(\mathbf{x})}{\mu}$ is the hydraulic conductivity tensor ($\text{m}^2/(\text{Pa} \cdot \text{s})$). To account for preferential bulk flow along white matter tracts, $\mathbf{k}(\mathbf{x})$ is constructed directly from patient-specific Diffusion Tensor Imaging (DTI) fractional anisotropy matrices:

$$\mathbf{k}(\mathbf{x}) = k_{\text{iso}} \left[(1 - \alpha_{\text{ani}})\mathbf{I} + \alpha_{\text{ani}}\mathbf{v}_1\mathbf{v}_1^T \right] \quad (3)$$

where $\mathbf{v}_1$ is the principal DTI tractography eigenvector and $\alpha_{\text{ani}} \in [0, 1]$ is a model parameter representing local directional anisotropy. The mapping from DTI signal to hydraulic permeability is a modeling assumption that requires calibration against experimental transport data.

Steady-state conservation of fluid mass within deformable porous brain tissue satisfies:

$$\nabla \cdot \mathbf{v} = S_{\text{infusion}} - L_p (p - p_{\text{microvasc}}) \quad (4)$$

where $S_{\text{infusion}}(\mathbf{x}, t)$ represents the local source term, L_p is an effective microvascular filtration coefficient, and $p_{\text{microvasc}}$ is an effective vascular pressure term. Numeric parameter values must be fitted to the specific experimental system.

For each therapeutic biologic and small molecule in the RVNA stack (SYN023 full-length IgG, biparatopic VHH nanobodies, BCX4430, custom ASOs, NT-3, and NGF), spatial and temporal concentration distributions $C_i(\mathbf{x}, t)$ are governed by the convection-dispersion-reaction transport equation:

$$\frac{\partial(\theta C_i)}{\partial t} + \nabla \cdot (\mathbf{v} C_i) = \nabla \cdot (\mathbf{D}_i^* \nabla C_i) - k_{\text{elim},i} C_i - k_{\text{on},i} C_i (B_{\text{max},i} - B_i) + k_{\text{off},i} B_i \quad (5)$$

where:

- θ is the extracellular volume fraction (θ is an experimentally or literature-calibrated extracellular volume fraction);
- $\mathbf{D}_i^* = \frac{D_{0,i}}{\lambda_{\text{tort}}^2} \mathbf{I} + \mathbf{D}_{\text{mech}}(\mathbf{v})$ is the effective hydrodynamic dispersion tensor incorporating tissue tortuosity (λ_{tort} is a calibrated tissue tortuosity parameter) and velocity-dependent mechanical dispersion;
- $k_{\text{elim},i}$ is the first-order clearance and enzymatic degradation rate;
- B_i is the concentration of drug-target complexes (e.g., antibody-virion or neurotrophin-receptor conjugates), with association rate $k_{\text{on},i}$ and dissociation rate $k_{\text{off},i}$.

Infusion-induced tissue displacement $\mathbf{u}$ and backflow risk along the catheter tract are modeled using Biot's poroelasticity formulation:

$$\nabla \cdot \boldsymbol{\sigma}' - \alpha \nabla p = \mathbf{0}, \quad \boldsymbol{\sigma}' = 2G\boldsymbol{\varepsilon} + \lambda \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} \quad (6)$$

where $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the strain tensor, α is a fitted Biot coefficient is the Biot coefficient, and G, λ are the Lamé elasticity moduli of brain tissue.

A future research release could provide:

1. **High-Performance CED Simulation Engine:** Open-source finite-volume/difference code implementing equations (2)–(6) for multi-drug CED co-infusion under the GNU General Public License v3.0. **Additionally, a SIMD-optimized C++ code is coming for all the core equations to provide massive computational speedups.**
2. **Automated V_d/V_i Predictive Optimization:** Algorithmic trajectory optimization that estimates distribution-to-infusion ratios under empirically calibrated tissue parameters rather than imposing a fixed rabies-specific range.
3. **Clinical DICOM/DTI Importer:** Automated pipeline transforming patient-specific MRI DICOM datasets into high-density 3D volumetric finite-element meshes.
4. **Full Source Code:** Available at `ced_simulation/` with publication-quality visualization output.

12 Conclusion

“Although preventive therapy is highly successful after recognized exposures, no established therapy reliably cures patients who develop rabies encephalomyelitis, which remains an important therapeutic challenge.” [Jackson, 2011]

For decades, management of symptomatic rabies has been constrained by the absence of a reproducibly effective disease-modifying therapy and by uncertainty about the benefit of aggressive experimental regimens. The definitive dismantlement of the Milwaukee Protocol — documented through > 64 clinical failures worldwide [Jackson, 2025, Jackson and Hemachudha, 2026] — leaves the field without any active alternative. RVNA proposes a research paradigm centered on testing active clearance and neurorepair hypotheses rather than assuming that passive sedation or any single experimental modality is sufficient.

By combining delivery science, antibody engineering, antiviral hypotheses, neuroprotection, and computational modeling, the RVNA framework provides a detailed agenda for translational investigation. It does not provide an established clinical treatment or an actionable bedside protocol.

“More basic research is needed to improve our understanding of the mechanisms involved in rabies pathogenesis, which will hopefully facilitate the development of novel therapeutic approaches in the future.” [Jackson, 2011]

The RVNA framework is offered as a hypothesis-generating research document whose claims should be revised as new evidence becomes available.

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Medical and Legal Disclaimer

This document describes a theoretical biomedical systems engineering proposal for scientific research and expanded-access evaluation. It does not constitute formal medical advice. All clinical procedures, expanded-access applications, and surgical interventions must be evaluated and administered by licensed medical specialists in accordance with applicable national laws, institutional review boards (IRBs), and regulatory health authorities.

References

Heather M. Callaway, Dawid Zyla, Florence Larrous, Guilherme Dias de Melo, Kathryn M. Hastie, Ruben Diaz Avalos, Alyssa Agarwal, Davide Corti, Hervé Bourhy, and Erica

- Ollmann Saphire. Structure of the rabies virus glycoprotein trimer bound to a prefusion-specific neutralizing antibody. *Science Advances*, 8(24):eabp9151, 2022. doi: 10.1126/sciadv.abp9151. URL <https://doi.org/10.1126/sciadv.abp9151>.
- Centers for Disease Control and Prevention. Rabies post-exposure prophylaxis guidance. <https://www.cdc.gov/rabies/hcp/clinical-care/post-exposure-prophylaxis.html>, 2026a. Updated 11 August 2026.
- Centers for Disease Control and Prevention. About rabies. <https://www.cdc.gov/rabies/about/index.html>, 2026b. Updated 27 January 2026.
- Thanathom Chailangkarn, Nathiphat Tanwattana, Thanakorn Jaemthaworn, Sira Sriswasdi, Nanchaya Wanasen, Sithichoke Tangphatsornruang, Kantinan Leetanasakul, Yuparat Jantraphakorn, Wanapinun Nawae, and Challika Kaewborisuth. Establishment of human-induced pluripotent stem cell-derived neurons—a promising in vitro model for a molecular study of rabies virus and host interaction. *International Journal of Molecular Sciences*, 22(21):11986, 2021. doi: 10.3390/ijms222111986. URL <https://doi.org/10.3390/ijms222111986>.
- Anthony R. Fooks and Alan C. Jackson, editors. *Rabies: Scientific Basis of the Disease and Its Management*. Academic Press, 4th edition, 2020. doi: 10.1016/C2017-0-04754-0. URL <https://doi.org/10.1016/C2017-0-04754-0>.
- Zhen F. Fu, X. Li, and V. Dhingra. Pathogenic rabies virus alters host protein expression in the central nervous system: implications for neuronal dysfunction. *Developments in Biologicals*, 131:83–91, 2008. URL <https://pubmed.ncbi.nlm.nih.gov/18634469/>.
- Shani Gluska, Eitan Erez Zahavi, Michael Chein, Tal Gradus, Anja Bauer, Stefan Finke, and Eran Perlson. Rabies virus hijacks and accelerates the p75^{ntr} retrograde axonal transport machinery. *PLoS Pathogens*, 10(8):e1004348, 2014. doi: 10.1371/journal.ppat.1004348. URL <https://doi.org/10.1371/journal.ppat.1004348>.
- Lacey W. Heinsberg, Aboli Kesbhat, Bailey Petersen, Lauren Kaseman, Zachary Stec, Nivinthiga Anton, Patrick M. Kochanek, Keith Owen Yeates, Daniel E. Weeks, Yvette P. Conley, and Amery Treble-Barna. Differential dna methylation of the brain-derived neurotrophic factor gene is observed after pediatric traumatic brain injury compared with orthopedic injury. *Journal of Neurotrauma*, 43(7–8):575–584, 2026. doi: 10.1177/08977151251400737. URL <https://doi.org/10.1177/08977151251400737>.
- Steven A. Hill, Isabel Bravo-Ferrer, Austėja Čiulkinytė, Noelia Pérez Ramos, Ilaria Rossetti, et al. Molecular profiling of brain endothelial cell to astrocyte endfoot communication in mouse and human. *Nature Communications*, 16:9750, 2025. doi: 10.1038/s41467-025-65487-4. URL <https://doi.org/10.1038/s41467-025-65487-4>.
- M. Iwata, S. Komori, T. Unno, N. Minamoto, and H. Ohashi. Modification of membrane currents in mouse neuroblastoma cells following infection with rabies virus. *British Journal of Pharmacology*, 126(8):1691–1698, 1999. doi: 10.1038/sj.bjp.0702473. URL <https://doi.org/10.1038/sj.bjp.0702473>.
- Alan C. Jackson. Therapy of human rabies. In *Advances in Virus Research*, volume 79, pages 365–375. Academic Press, 2011. doi: 10.1016/B978-0-12-387040-7.00017-2. URL <https://doi.org/10.1016/B978-0-12-387040-7.00017-2>.

- Alan C. Jackson. Demise of the milwaukee protocol for rabies. *Clinical Infectious Diseases*, 81(4):e229–e232, 2025. doi: 10.1093/cid/ciaf157. URL <https://doi.org/10.1093/cid/ciaf157>.
- Alan C. Jackson and Anthony R. Fooks, editors. *Rabies: Scientific Basis of the Disease and Its Management*. Elsevier, 5th edition, 2025. ISBN 9780443299797. URL <https://shop.elsevier.com/books/rabies/jackson/978-0-443-29979-7>.
- Alan C. Jackson and Zhen F. Fu. Pathogenesis. In *Rabies: Scientific Basis of the Disease and Its Management*, chapter 8, pages 299–349. Academic Press, 3rd edition, 2013. doi: 10.1016/B978-0-12-396547-9.00008-0. URL <https://doi.org/10.1016/B978-0-12-396547-9.00008-0>.
- Alan C. Jackson and Thiravat Hemachudha. Reply to willoughby on the demise of the milwaukee protocol for rabies. *Clinical Infectious Diseases*, 82(5):e1126–e1127, 2026. doi: 10.1093/cid/ciaf734. URL <https://doi.org/10.1093/cid/ciaf734>.
- Marian Lacy, Nonthapan Phasuk, and Stephen J. Scholand. Human rabies treatment—from palliation to promise. *Viruses*, 16(1):160, 2024. doi: 10.3390/v16010160. URL <https://doi.org/10.3390/v16010160>.
- Xiaoqiang Liu, Jingyu Li, Yongxian Zha, Zhengxiong Wang, Ya Jiang, Xiangyu Zhang, Jiangshu Guo, Jiajie Yu, Xiangjun Li, Qingchao Zhang, Caroline Reid, J. Bruce McClain, and Eric Tsao. The efficacy and safety of syn023 (zamerovimab and mazorelvimab injection), the recombinant humanized anti-rabies virus monoclonal antibody mixture, combined with rabies vaccine in a who category iii rabies post-exposure population: A randomized, double-blind, positive control, phase iii clinical trial. *Vaccine*, 61:127289, 2025. doi: 10.1016/j.vaccine.2025.127289. URL <https://doi.org/10.1016/j.vaccine.2025.127289>.
- Peter F. Morgenstern, Zhiping Zhou, Eva Wembacher-Schröder, Vincent Cina, Apostolos John Tsiouris, and Mark M. Souweidane. Clinical tolerance of corticospinal tracts in convection-enhanced delivery to the brainstem. *Journal of Neurosurgery*, 131(6):1812–1818, 2019. doi: 10.3171/2018.6.JNS18854. URL <https://doi.org/10.3171/2018.6.JNS18854>.
- Vikas Munjal, Asad S. Akhter, Matthew T. Rocco, Massimo S. Fiandaca, John D. Heiss, and Krystof S. Bankiewicz. Bilateral putaminal convection of adeno-associated virus vector serotype 2 glial cell line-derived neurotrophic factor gene therapy in parkinson’s disease may provide glial cell line-derived neurotrophic factor transgene expression to caudate nucleus: Potential role of perivascular and other low resistance leakage pathways. *Neurosurgery*, 98(6):1241–1246, 2026. doi: 10.1227/neu.00000000000003729. URL <https://doi.org/10.1227/neu.00000000000003729>.
- Kazim H. Narsinh, Karishma Kumar, Krystof Bankiewicz, Alastair J. Martin, Mitchell Berger, Jennifer Clarke, Jennie Taylor, Nancy Ann Oberheim Bush, Annette M. Molinaro, Manish Aghi, and Nicholas Butowski. A phase i study of convection-enhanced delivery (ced) of liposomal-irinotecan using real-time magnetic resonance imaging in patients with recurrent high-grade glioma. *Journal of Neuro-Oncology*, 172(1):219–227, 2025. doi: 10.1007/s11060-024-04904-y. URL <https://doi.org/10.1007/s11060-024-04904-y>.

- Sara Jamshidi Parvar, Cheng In Wong, Andrew R. Lewis, Elwira Szychot, Christopher J. Morris, David Shorthouse, and Karolina Dziemidowicz. Convection-enhanced delivery for brain malignancies: Technical parameters, formulation strategies and clinical perspectives. *Advanced Drug Delivery Reviews*, 224:115657, 2025. doi: 10.1016/j.addr.2025.115657. URL <https://doi.org/10.1016/j.addr.2025.115657>.
- Meishen Ren, Jiaojiao Zhou, Zhiyong Song, Hong Mei, Ming Zhou, Zhen F. Fu, Heyou Han, and Ling Zhao. Aptamer and rvg functionalized gold nanorods for targeted photothermal therapy of neurotropic virus infection in the mouse brain. *Chemical Engineering Journal*, 411:128557, 2021. doi: 10.1016/j.cej.2021.128557. URL <https://doi.org/10.1016/j.cej.2021.128557>.
- Zeheng Ren, Caiqian Wang, Haoran Wang, Qiong Wu, Qingxiu Hou, Xue Qi, Wenna He, Xiaoyu Zhang, Jiawu Wan, Zhen F. Fu, Ming Zhou, and Ling Zhao. A synb1-conjugated antibody cocktail crosses the blood-brain barrier to produce a therapeutic effect on rabies. *Proceedings of the National Academy of Sciences of the United States of America*, 122(52):e2516465122, 2025. doi: 10.1073/pnas.2516465122. URL <https://doi.org/10.1073/pnas.2516465122>.
- Christophe Rousselle, Maria Smirnova, Philippe Clair, Jeanne-Marie Lefauconnier, Alain Chavanieu, Bernard Calas, Jean-Michel Scherrmann, and Jamal Tamsamani. Enhanced delivery of doxorubicin into the brain via a peptide-vector-mediated strategy: Saturation kinetics and specificity. *Journal of Pharmacology and Experimental Therapeutics*, 296(1):124–131, 2001. doi: 10.1016/S0022-3565(24)29672-5. URL [https://doi.org/10.1016/S0022-3565\(24\)29672-5](https://doi.org/10.1016/S0022-3565(24)29672-5).
- Courtney A. Scott, John P. Rossiter, R. David Andrew, and Alan C. Jackson. Structural abnormalities in neurons are sufficient to explain the clinical disease and fatal outcome of experimental rabies in yellow fluorescent protein-expressing transgenic mice. *Journal of Virology*, 82(1):513–521, 2008. doi: 10.1128/JVI.01677-07. URL <https://doi.org/10.1128/JVI.01677-07>.
- Susan E. Sloan, Cathleen Hanlon, William Weldon, Michael Niezgoda, Jesse Blanton, Josh Self, Kirk J. Rowley, Robert B. Mandell, Gregory J. Babcock, William D. Jr. Thomas, Charles E. Rupprecht, and Donna M. Ambrosino. Identification and characterization of a human monoclonal antibody that potently neutralizes a broad panel of rabies virus isolates. *Vaccine*, 25(15):2800–2810, 2007. doi: 10.1016/j.vaccine.2006.12.031. URL <https://doi.org/10.1016/j.vaccine.2006.12.031>.
- Yan Song, Jinli Hou, Bin Qiao, Yanchao Li, Ye Xu, Ming Duan, Zhenhong Guan, Maolin Zhang, and Liankun Sun. Street rabies virus causes dendritic injury and f-actin depolymerization in the hippocampus. *Journal of General Virology*, 94(2):276–283, 2013. doi: 10.1099/vir.0.047480-0. URL <https://doi.org/10.1099/vir.0.047480-0>.
- Sanne Terryn, Aurélie Francart, Heidi Rommelaere, Catelijne Stortelers, and Steven Van Gucht. Post-exposure treatment with anti-rabies vhh and vaccine significantly improves protection of mice from lethal rabies infection. *PLoS Neglected Tropical Diseases*, 10(8):e0004902, 2016. doi: 10.1371/journal.pntd.0004902. URL <https://doi.org/10.1371/journal.pntd.0004902>.

- Yasuo Uchida, Yuta Yagi, Masaki Takao, Mitsutoshi Tano, Mina Umetsu, Satoshi Hirano, Takuya Usui, Masanori Tachikawa, and Tetsuya Terasaki. Comparison of absolute protein abundances of transporters and receptors among blood–brain barriers at different cerebral regions and the blood–spinal cord barrier in humans and rats. *Molecular Pharmaceutics*, 17(6):2006–2020, 2020. doi: 10.1021/acs.molpharmaceut.0c00178. URL <https://doi.org/10.1021/acs.molpharmaceut.0c00178>.
- U.S. Food and Drug Administration. Expanded access to investigational drugs for treatment use: Questions and answers. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/expanded-access-investigational-drugs-treatment-use-questions-and-answers>, 2026. Final guidance revised April 2026.
- Jr. Willoughby, Rodney E., Kelly S. Tieves, George M. Hoffman, Nancy S. Ghanayem, Catherine M. Amlie-Lefond, Michael J. Schwabe, Michael J. Chusid, and Charles E. Rupprecht. Survival after treatment of rabies with induction of coma. *New England Journal of Medicine*, 352(24):2508–2514, 2005. doi: 10.1056/NEJMoa050382. URL <https://doi.org/10.1056/NEJMoa050382>.
- World Health Organization. Rabies. <https://www.who.int/news-room/fact-sheets/detail/rabies>, 2024. Fact sheet updated 5 June 2024.
- Yuan Xie, Ying-Lin Chi, Shu-Qing Liu, and Wu-Yang Zhu. Bcx4430 inhibits the replication of rabies virus by suppressing mtor-dependent autophagy in vitro. *Virology*, 585:21–31, 2023. doi: 10.1016/j.virol.2023.05.012. URL <https://doi.org/10.1016/j.virol.2023.05.012>.
- Xiong Xu, Lili Song, Yueying Li, Jin Guo, Shuo Huang, Shuang Du, Weizhen Li, Rangjuan Cao, and Shusen Cui. Neurotrophin-3 promotes peripheral nerve regeneration by maintaining a repair state of schwann cells after chronic denervation via the trkc/erk/c-jun pathway. *Journal of Translational Medicine*, 21:733, 2023. doi: 10.1186/s12967-023-04609-2. URL <https://doi.org/10.1186/s12967-023-04609-2>.
- Na Yan, Bohan Hu, Huina Zhai, Xu Han, Cuiling Hu, Xueyi Guan, and Jian Gong. Structural and functional alterations in the contralateral hemisphere following pediatric intracranial surgery: a pilot longitudinal neuroimaging study. *Frontiers in Human Neuroscience*, 19:1568945, 2025. doi: 10.3389/fnhum.2025.1568945. URL <https://doi.org/10.3389/fnhum.2025.1568945>.
- Jianqing Zhao, Zonghui Zeng, Yixi Chen, Wei Liu, Huanchun Chen, Zhen F. Fu, Ling Zhao, and Ming Zhou. Lipid droplets are beneficial for rabies virus replication by facilitating viral budding. *Journal of Virology*, 96(2):e01473–21, 2022. doi: 10.1128/JVI.01473-21. URL <https://doi.org/10.1128/JVI.01473-21>.